CARE TRANSITION EFFICIENCY & PLACEMENT OUTCOME ANALYTICS

Research Paper | Unified Mentor Project

Data-driven exploratory analysis of CBP custody, transfers to HHS care, and HHS discharges

Prepared from: Jupyter Notebook (Untitled.ipynb), project CSV dataset, and implemented Streamlit dashboard

Analysis period: January 12, 2023 – December 21, 2025

Prepared for project submission

Abstract

This project examines reported daily activity across the U.S. Unaccompanied Alien Children (UAC) program pipeline, focusing on children apprehended and placed in Customs and Border Protection (CBP) custody, children in CBP custody, transfers out of CBP custody, children in Department of Health and Human Services (HHS) care, and discharges from HHS care. The original CSV contains 1,170 rows and six fields. After removing 450 completely blank rows and parsing dates and numeric fields, 720 valid reporting records remained. Exploratory analysis includes descriptive statistics, time-series trends, transfer and discharge ratios, weekday/weekend comparisons, entry–exit balance, and changes in reported custody/care counts. A Streamlit dashboard was built to present aggregate indicators, trends, and threshold-based alerts. The analysis is descriptive and operationally exploratory; the dataset does not link individual children across stages, so aggregate ratios should not be interpreted as individual-level outcomes or cohort conversion rates.

Keywords

Data analytics; exploratory data analysis; CBP; HHS care; transfer efficiency; discharge; Streamlit dashboard; operational flow

1. Introduction

Public-sector care processes involve multiple stages, reporting measures, and handoffs. Monitoring the reported counts at each stage can help analysts identify changes in workload, transfer activity, and discharge activity over time. This project uses a provided daily dataset to explore these measures and build a dashboard that makes the results easier to inspect.

The project's objective is to develop an analytical view of the care pipeline and identify descriptive patterns that may warrant operational review. The work combines data cleaning, exploratory data analysis (EDA), ratio-based indicators, time-series visualization, and a Streamlit dashboard.

The analysis does not estimate causal effects, individual waiting times, or outcomes for a matched group of children. The available fields are aggregate reporting measures and should be interpreted within that limitation.

2. Objectives

The main objectives are: (1) clean and validate the supplied dataset; (2) describe its reporting period and distributions; (3) analyze CBP custody, transfers, HHS care, and discharges over time; (4) calculate the project's operational ratios and flow-balance indicators; (5) identify dates with large reported changes; and (6) present the results through a user-friendly Streamlit dashboard.

3. Dataset Description

The dataset file is HHS_Unaccompanied_Alien_Children_Program.csv. It contains six columns: Date; Children apprehended and placed in CBP custody; Children in CBP custody; Children transferred out of CBP custody; Children in HHS Care; and Children discharged from HHS Care. The original file contains 1,170 rows, including blank rows. After cleaning, the analysis contains 720 dated records spanning January 12, 2023 through December 21, 2025.

The reporting dates are not a complete daily calendar series. Therefore, "daily" in this paper refers to each available reporting date, not necessarily every calendar day. The records are sorted chronologically for trend analysis.

Dataset attribute	Result
Original rows	1,170
Completely blank rows	450

Valid dated records	720
Fields	13
Date coverage	12 Jan 2023 – 21 Dec 2025
Missing values after cleaning	2
Duplicate rows after cleaning	0

4. Data Cleaning and Preprocessing

The notebook applies the following preparation steps:

• Loaded the CSV file into a pandas DataFrame.
• Inspected shape, column names, data types, missing values, duplicate rows, and sample records.
• Removed fully blank rows using dropna(how='all').
• Converted Date to datetime format and sorted records chronologically.
• Converted numeric columns to numeric values; comma separators were removed from the HHS care field before conversion.
• Rechecked the cleaned dataset for missing values and duplicates.

The original duplicate count was dominated by repeated completely blank rows. After dropping fully blank rows, the cleaned data contains no duplicate rows and no missing values in the six analysis columns. This distinction is important: the 449 duplicates reported before cleaning should not be treated as duplicates among actual reporting records.

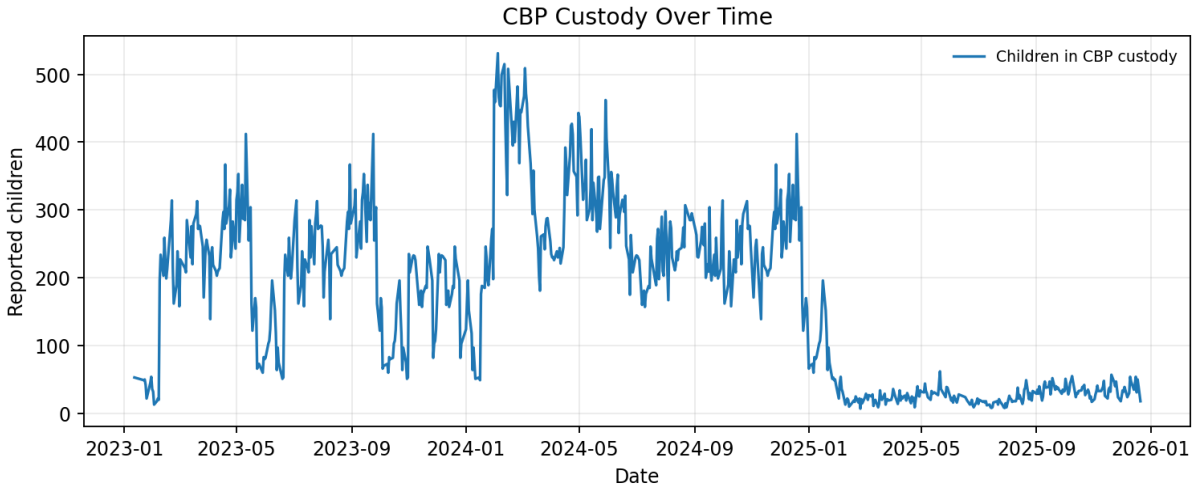
5. Exploratory Data Analysis

5.1 Descriptive Statistics

The descriptive statistics show substantial variation across reporting dates. CBP custody, transfer counts, HHS care, and discharge counts fluctuate over the period. The following selected values summarize the cleaned dataset:

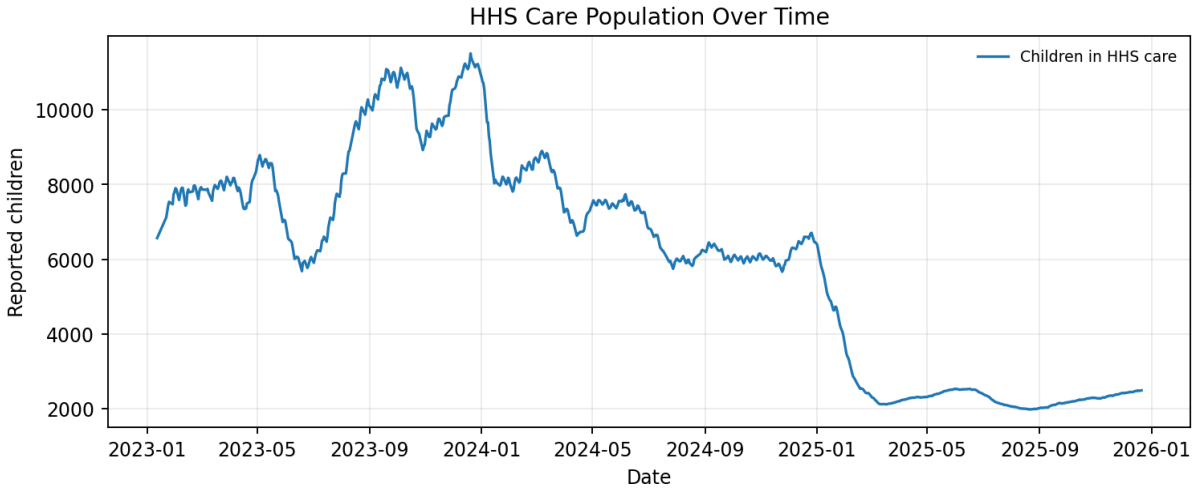
Measure	Mean	Minimum	Maximum
Apprehensions / CBP placement	93.52	0.00	333.00
Children in CBP custody	171.49	7.00	531.00
CBP transfers	128.67	0.00	440.00
Children in HHS care	6,061.27	1,972.00	11,516.00
HHS discharges	173.41	0.00	505.00

5.2 CBP Custody Trend



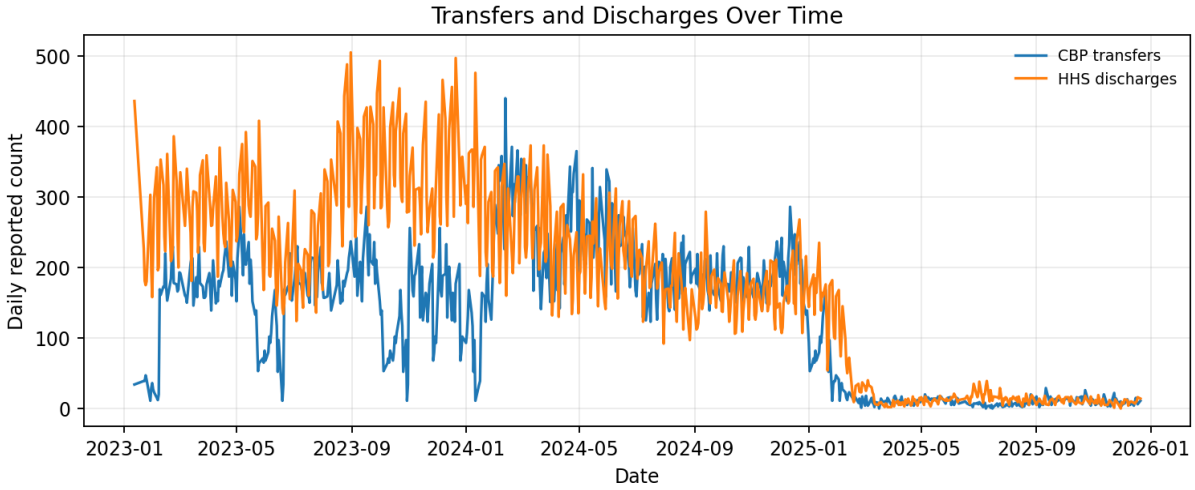
The CBP custody series varies over time rather than following a smooth linear trajectory. Peaks and changes should be reviewed alongside the underlying reporting context; this dataset alone cannot explain why a particular increase or decrease occurred.

5.3 HHS Care Population Trend



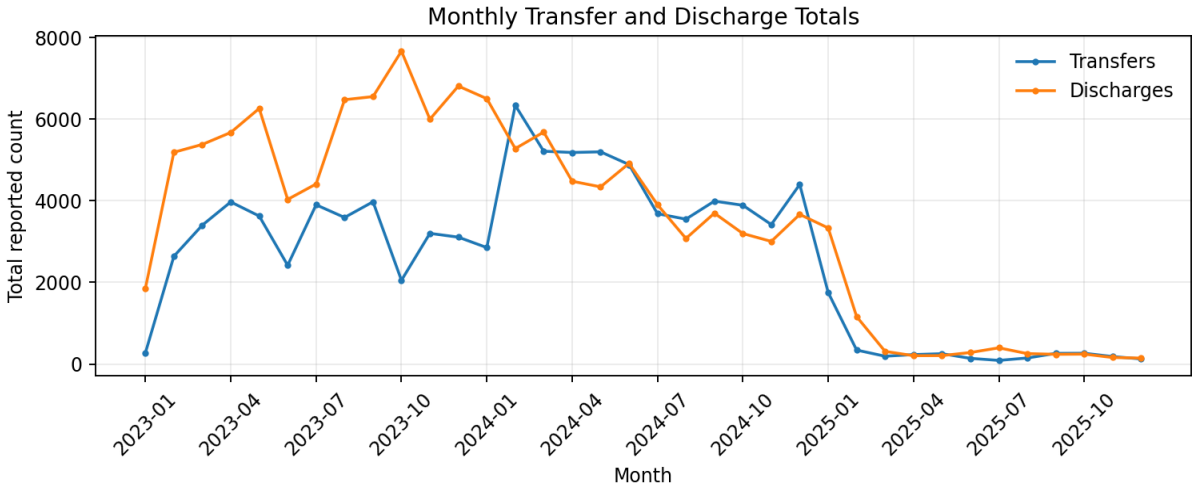
The HHS care population also fluctuates. The graph provides a descriptive view of the reported population size, not a measure of individual duration in care.

5.4 Transfers and Discharges



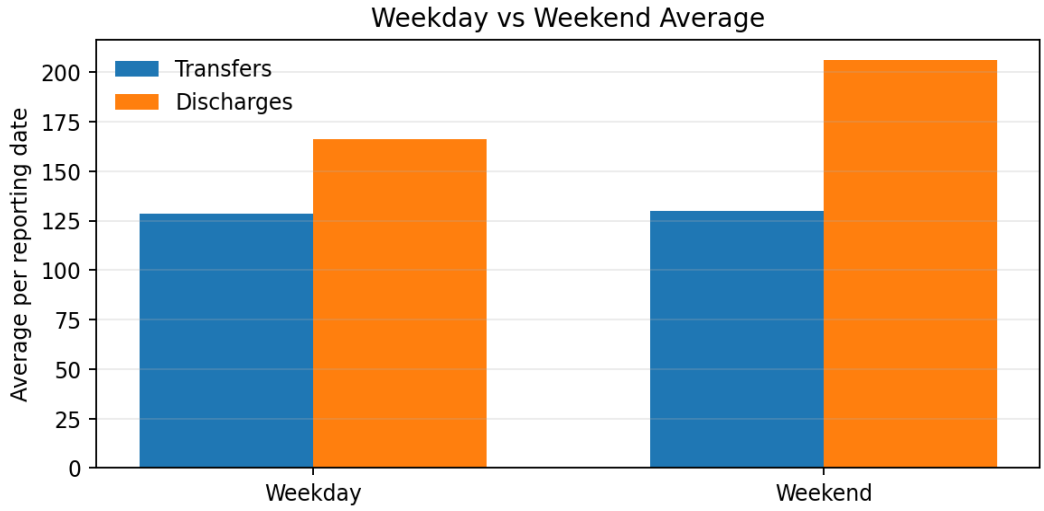
Transfers and discharges are both flow measures. They vary considerably by reporting date and should be compared with stock measures cautiously because the counts are not linked at the individual level.

5.5 Monthly Transfer and Discharge Trends



Monthly totals aggregate the available reporting records by calendar month. The chart is useful for spotting periods with comparatively high or low activity, but the number of reporting records may vary by month and should be considered when comparing totals.

5.6 Weekday and Weekend Comparison



Day type	Average transfers	Average discharges
Weekday	128.36	166.19
Weekend	130.06	206.14

The dataset shows average transfer counts of 128.36 on weekdays and 130.06 on weekends. Average discharges are 166.19 on weekdays and 206.14 on weekends. This is a descriptive comparison only; it does not establish that day type causes the difference.

6. KPI Definitions and Results

The notebook uses ratios to summarize reporting-period activity. Because the dataset combines stock counts (children in custody/care) with flow counts (transfers/discharges), these ratios are operational indicators rather than conventional cohort-based rates. The numerator and denominator should be explicitly stated whenever the KPIs are presented.

6.1 Transfer Efficiency

Formula: total reported CBP transfers ÷ total reported children in CBP custody × 100. Using the cleaned dataset, this aggregate ratio is **75.03%**. It is a ratio of summed reporting values, not the average of daily percentages and not a probability that a specific child was transferred.

6.2 Discharge Effectiveness

Formula: total reported HHS discharges ÷ total reported children in HHS care × 100. The aggregate ratio is **2.86%**. As with transfer efficiency, this compares aggregated flow and stock measures and should not be interpreted as an individual discharge probability.

6.3 Pipeline Throughput (Notebook Definition)

The notebook defines total exits as CBP transfers plus HHS discharges, and calculates throughput as total exits ÷ total apprehensions/placements × 100. Under this definition, the result is **322.99%**. Since the numerator includes two different stage flows and is not a matched cohort, a value above 100% can occur and does not mean more than 100% of the same children exited.

6.4 Outcome Stability Score (Notebook Definition)

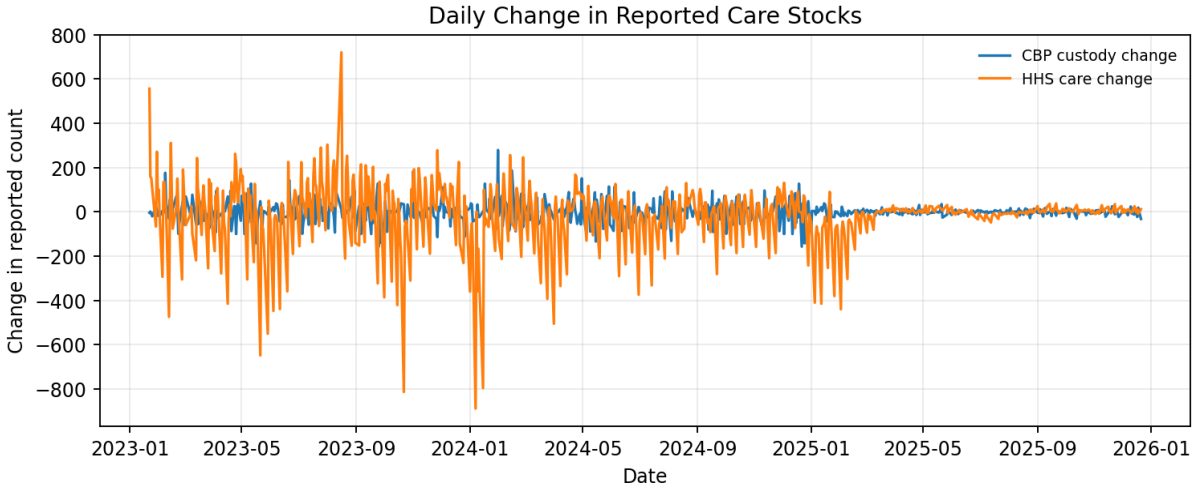
The notebook calculates the coefficient of variation (CV) of HHS discharges as standard deviation divided by mean, then uses Stability Score = 100 ÷ (1 + CV). The mean discharge count is 173.41, standard deviation is 125.70, CV is 0.725, and the notebook score is **57.97/100**. This is a custom variability-based score; higher values indicate less relative variation under this formula. It is not a validated program outcome measure.

7. Flow Balance and Potential Bottlenecks

The notebook defines total entries as apprehensions/placements and total exits as transfers plus discharges. Entry–exit difference is entries minus exits. A negative difference indicates that the combined reported exit flows exceed the reported entry flow for that date. This does not imply the system is clearing the same cohort because reporting categories may represent different populations and stages.

Across the cleaned dataset, average entries are 93.52 per reporting record and average combined exits are 302.07. The average entry-minus-exit difference is -208.55. The large negative average should be treated as a signal to examine definitions, time alignment, and stock-versus-flow interpretation—not as evidence of a negative population.

The notebook also calculates changes in CBP custody and HHS care using the difference between consecutive reporting records. Positive changes indicate an increase in the reported stock relative to the previous available record; negative changes indicate a decrease. These changes are not waiting-time estimates.



The largest observed positive CBP stock change in the notebook is 279 on January 31, 2024. The largest observed positive HHS stock change is 720 on August 16, 2023. These are candidates for contextual review, not confirmed bottlenecks or explanations of why the counts changed.

8. Streamlit Dashboard

The Streamlit dashboard operationalizes the analysis and includes the project's four required modules:

1. Care Pipeline Flow Visualization: an aggregate Sankey-style diagram that illustrates reported stages. The visualization is explicitly illustrative and is not a matched cohort flow.

2. Transfer & Discharge Efficiency: KPI cards and a ratio chart for transfer efficiency, discharge effectiveness, and a throughput proxy, with metric toggles.

3. Bottleneck Detection: CBP and HHS net-flow series, a configurable threshold, and reporting dates that exceed the selected threshold.

4. Outcome Trend Analysis: daily/monthly discharge and transfer trends, including weekday/weekend comparison.

The dashboard also provides date filtering and filtered-data download. During the user's dashboard check, 720 records were loaded and 23 reporting dates were flagged at the current net-flow threshold of 100. The alert count is sensitive to the selected date range and threshold.

9. KPI Consistency Note

The current dashboard and notebook contain different definitions for two proxy KPIs. This should be resolved before final submission:

Indicator	Notebook calculation	Current dashboard calculation
Pipeline throughput	(Transfers + discharges) ÷ apprehensions × 100 = 322.99%	Discharges ÷ apprehensions × 100 = 185.42%
Outcome stability	100 ÷ (1 + CV) = 57.97/100	100 × (1 – CV), bounded 0–100 = 27.51/100

Recommendation: For a consistent submission, update the dashboard formulas to match the notebook or explicitly rename and document the dashboard alternatives. The values should not be mixed in the same paper without this explanation.

10. Key Findings

• The dataset was reduced from 1,170 raw rows to 720 valid dated records after removing fully blank rows and cleaning data types.
• The valid records span January 12, 2023 to December 21, 2025, but are not a complete daily calendar series.
• Transfer and discharge activity show considerable variation over time, supporting the use of time-series monitoring rather than relying on a single average.
• Average weekend discharges are higher than weekday discharges in this dataset, while average transfer counts are similar across day types. This descriptive pattern requires context before operational interpretation.
• The aggregate transfer ratio is 75.03% and aggregate discharge ratio is 2.86% under the stated sum-based formulas.
• The notebook's throughput ratio is 322.99% because total exits includes transfers and discharges while apprehensions are the denominator. It is a non-cohort aggregate ratio, not a direct conversion rate.
• Reported stock changes can identify dates for review, but do not establish causes or actual waiting times.
• The dashboard is useful for monitoring, provided its KPI definitions are aligned with the notebook.

11. Recommendations

1. Standardize KPI definitions across the notebook, dashboard, and report. Display the exact numerator and denominator next to each KPI.
2. Avoid labeling stock-to-flow ratios as individual-level efficiency or probability. Prefer labels such as “aggregate transfer-to-custody ratio” where appropriate.
3. Validate reporting definitions and timing with the dataset owner, especially when combining transfers and discharges into a single exit measure.
4. Add a data completeness view showing the number of reporting records per month and gaps in the calendar series.
5. Investigate the largest positive and negative stock changes with source documentation before assigning operational causes.
6. Keep the threshold alert configurable and describe it as a screening alert rather than a confirmed bottleneck.
7. Add a dashboard note explaining that records are aggregate and do not track individual children across stages.
8. For future analysis, obtain stage-linked cohort data, consistent date definitions, and individual-level timestamps only if authorized and ethically appropriate. Such data would be required to estimate transition times or cohort-based outcomes.

12. Limitations

This analysis has several limitations. First, the dataset contains aggregate counts and does not identify individuals or link records across stages. Second, the reporting dates are irregular rather than daily for every calendar date. Third, the analysis is descriptive and cannot establish causality. Fourth, flow and stock variables may have different definitions and reporting windows, so ratios can be difficult to interpret. Fifth, dashboard KPI formulas currently differ from the notebook for throughput and stability; these should be aligned before the final project presentation. Finally, the threshold alert is a configurable heuristic and not a statistically validated bottleneck detector.

13. Conclusion

This project demonstrates a complete analytics workflow—from CSV inspection and cleaning to EDA, KPI calculation, visualization, and dashboard development—for examining reported CBP custody, transfers, HHS care, and discharges. The cleaned dataset contains 720 reporting records across nearly three years. The analysis identifies temporal variation, weekday/weekend differences, aggregate flow ratios, and dates with notable changes in reported stocks. The Streamlit dashboard makes these results accessible through interactive filters, charts, and threshold alerts.

The most important next step is methodological consistency: clearly distinguish aggregate ratios from cohort-based measures and align KPI formulas across the notebook and dashboard. With those definitions and data limitations made explicit, the project provides a useful descriptive monitoring prototype and a foundation for more rigorous process analysis.

Appendix A. KPI Formula Summary

KPI	Formula used in notebook	Interpretation
Transfer efficiency	Σ transfers ÷ Σ CBP custody × 100	Aggregate flow-to-stock ratio
Discharge effectiveness	Σ HHS discharges ÷ Σ HHS care × 100	Aggregate flow-to-stock ratio
Pipeline throughput	Σ (transfers + discharges) ÷ Σ apprehensions × 100	Aggregate exit/entry proxy; not cohort conversion
Outcome stability	100 ÷ (1 + discharge CV)	Custom relative-variability score
CBP net/stock change	Current CBP custody – previous record	Change between adjacent reporting records
HHS net/stock change	Current HHS care – previous record	Change between adjacent reporting records

Appendix B. Software and Tools

Python; pandas; NumPy; Matplotlib; Jupyter Notebook; Streamlit; Plotly. The notebook performs data preparation and exploratory analysis, while Streamlit presents the interactive dashboard.